MATH 999: Introduction to Widget Theory

Chapter 1: Foundations of Widget Theory

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1 Basic Definitions

We need a way to measure the size of a widget that captures both its width and depth.

Definition: Widget Norm

Let W be a widget space. For $w \in W$, the **widget norm** is

$$\|w\|_W = \sup_{k \geq 1} |w_k| \cdot 2^{-k}.$$

We say w is **finite** if $\|w\|_W < \infty$.

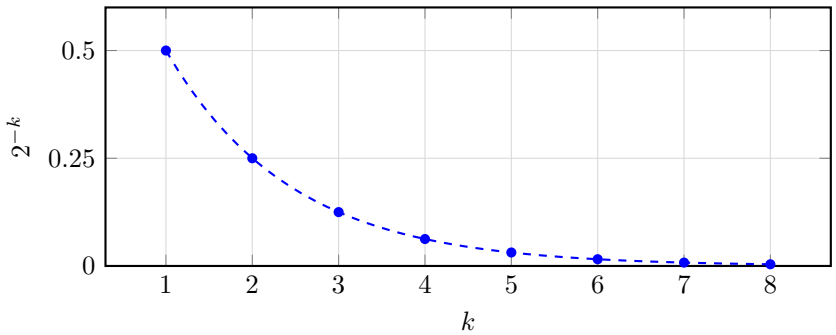
The factor 2^{-k} ensures later components contribute less.

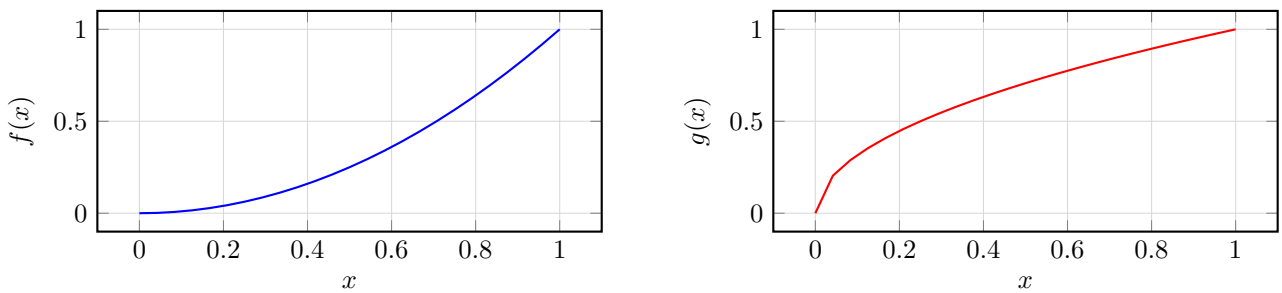
Example: Computing the widget norm

Let $w = (3, 1, 4, 1, 5, 9, \ldots)$ where w_k is the k-th digit of π.
The sup is attained at $k = 1$ because the exponential decay 2^{-k} dominates for all later terms.

$$\|w\|_W = \sup\left\{\frac{3}{2}, \frac{1}{4}, \frac{4}{8}, \frac{1}{16}, \frac{5}{32}, \ldots\right\} = \frac{3}{2}.$$

Here is a plot of the decay 2^{-k} showing why later components are negligible:





(a) $f(x) = x^2$ (b) $g(x) = \sqrt{x}$

Figure 1: Two functions on $[0, 1]$: x^2 is convex, $\sqrt{x}$ is concave.

Section Summary

- The widget norm $\|w\|_W = \sup_k |w_k| \cdot 2^{-k}$ measures size with exponential decay.
- $\|w\|_W = 0$ iff $w = 0$. The norm satisfies the triangle inequality.
- Later components contribute exponentially less, which is the key to all convergence arguments.

2 The Main Theorem

Every bounded widget sequence has a convergent subsequence.

Theorem: Widget Approximation Theorem

Let $(w^{(n)})$ be a sequence of widgets with $\|w^{(n)}\|_W \leq M$ for all n. Then there exists a subsequence $(w^{(n_j)})$ and a widget $w^* \in W$ such that $\|w^{(n_j)} - w^*\|_W \to 0$ as $j \to \infty$.

Proof. We construct w^* component by component using a diagonal argument.

Step 1. $(w_1^{(n)})$ is bounded by $2M$. By Bolzano-Weierstrass, extract a subsequence with $w_1^{(n_j^1)} \to w_1^*$.

Step 2. From that subsequence, extract a further one with $w_2^{(n_j^2)} \to w_2^*$.

Step 3. Continue for all k. The diagonal subsequence $w^{(n_j^j)}$ converges in every component.

Step 4. Let $\varepsilon > 0$. Choose K with $M \cdot 2^{-K} < \varepsilon/2$. For j large enough:

$$\|w^{(n_j^j)} - w^*\|_W \leq \max\left(\max_{k \leq K} |w_k^{(n_j^j)} - w_k^*| \cdot 2^{-k},\ M \cdot 2^{-K}\right) < \varepsilon.$$

We showed that every bounded widget sequence has a norm-convergent subsequence. □

Intuition

Two ideas: Bolzano-Weierstrass applied infinitely many times (once per component) plus diagonal extraction, and the 2^{-k} decay meaning we only need finitely many components to be close.

The diagonal argument can be visualized as a table where we select entries along the diagonal:

	$k=1$	$k=2$	$k=3$	$k=4$
Subseq 1	w_1^*	·	·	·
Subseq 2	·	w_2^*	·	·
Subseq 3	·	·	w_3^*	·

3 A Useful Lemma

Lemma: Tail Bound

If $\|w\|_W \leq M$, then for all $K \geq 1$: $\sup_{k>K} |w_k| \cdot 2^{-k} \leq M$.

Proof. $\sup_{k>K} |w_k| \cdot 2^{-k}$ is the supremum over a subset of $\{|w_k| \cdot 2^{-k} : k \geq 1\}$, so it is at most the full supremum, which is $\|w\|_W \leq M$. □

Intuition

The tail of a widget (components $k > K$) contributes at most M to the norm. For the diagonal argument, this means we can ignore the tail once K is large enough.

4 Showcase: All Available Tools

Every tool the agent can use, demonstrated below.

4.1 Auto-width Table (tabularx)

Property	Widget Norm	Sup Norm
Decay weighting	Yes (2^{-k})	No
Tail control	Automatic	Manual
Useful for	Widget spaces	Function spaces

4.2 Code Listing (codebox)

Python: Computing the widget norm

```
def widget_norm(w):
    """Compute ||w||_W = sup |w_k| * 2^(-k)."""
    return max(abs(w[k]) * 2**(-(k+1))
               for k in range(len(w)))

# Example
w = [3, 1, 4, 1, 5, 9]
print(widget_norm(w))  # 1.5
```

4.3 Algorithm (algobox)

Algorithm: Widget Approximation via Diagonal Extraction

Input: Bounded sequence $(w^{(n)})$ with $\|w^{(n)}\|_W \leq M$
Output: Convergent subsequence $(w^{(n_j)})$

1. Set $S_0 = \{1, 2, 3, \ldots\}$
2. For $k = 1, 2, 3, \ldots$:
   (a) Extract $S_k \subseteq S_{k-1}$ such that $w_k^{(n)} \to w_k^*$ along S_k
3. Set $n_j =$ the j-th element of S_j (diagonal selection)
4. Return $(w^{(n_j)})$

4.4 Inline Note Demo

The widget norm is the foundation of everything that follows.

Controlling $\|w^{(n)} - w^*\|_W$ is the goal of every convergence argument in this chapter.

Every convergence argument in this chapter reduces to bounding this norm.

4.5 Long Table Demo (longtable)

k	$\lvert w_k\rvert \cdot 2^{-k}$	Running sup
1	$3/2 = 1.500$	1.500
2	$1/4 = 0.250$	1.500
3	$4/8 = 0.500$	1.500
4	$1/16 = 0.063$	1.500
5	$5/32 = 0.156$	1.500
6	$9/64 = 0.141$	1.500

Table 1: Step-by-step computation of $\|w\|_W$ for $w = (3, 1, 4, 1, 5, 9, \ldots)$.